# The C-index Multiverse

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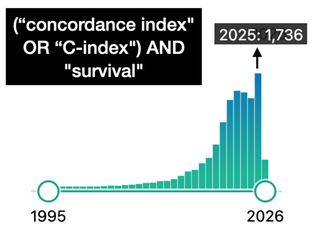
## Motivation

- **Quantifying out-of-sample discrimination performance** for time-to-event outcomes is a fundamental step for **model evaluation and selection** in the context of predictive modelling. The **C-index** is a widely used metric for this purpose, particularly with the growing development of machine learning methods.




- The **C-index** quantifies a model's ability to correctly **rank individuals** based on their predicted risk and observed time-to-event outcomes, initially proposed by Harrell and others (1982).

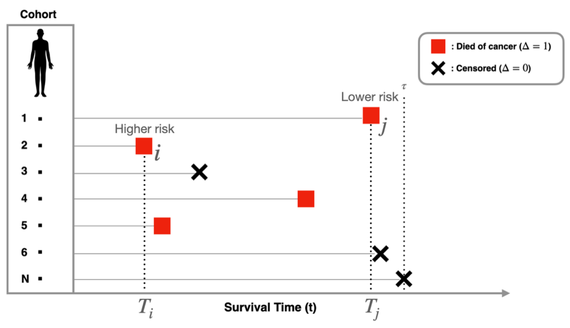
**How well is the model distinguishing between risk profiles?**

Model has good discrimination: $M(X_i) > M(X_j)$

Model has bad discrimination: $M(X_i) < M(X_j)$

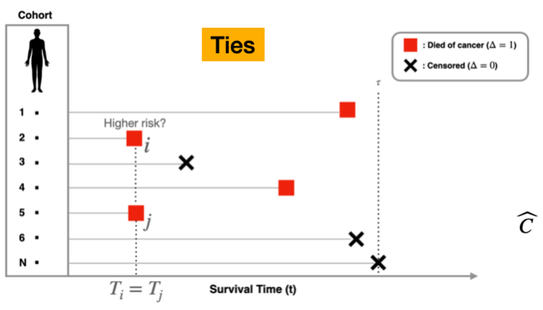
$$C = P(M(\mathbf{x}_i) > M(\mathbf{x}_j) \mid T_i < T_j)$$

**1.0** = perfect ranking, **0.5** ≈ random, **<0.5** = systematically wrong (inverted)




- **Problem:** There are **conceptually different C-index definitions and implementations** in R and Python, where **seemingly equal implementations can yield different results**. This can undermine reproducibility and complicate fair comparisons across models and studies.

## Ambiguity of in time-to-event data for C-index calculation

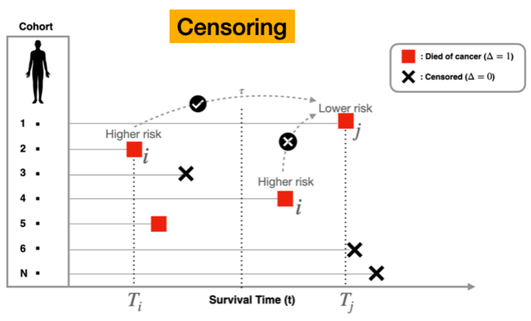



Ties in predictions

$$M(X_i) = M(X_j)$$

Informal definition

$$\widehat{C} = \frac{NumberConcordantPairs + 0.5(NumberTiedPredictionPairs)}{NumberComparablePairs}$$




Concordance probability (Uno's 2011)

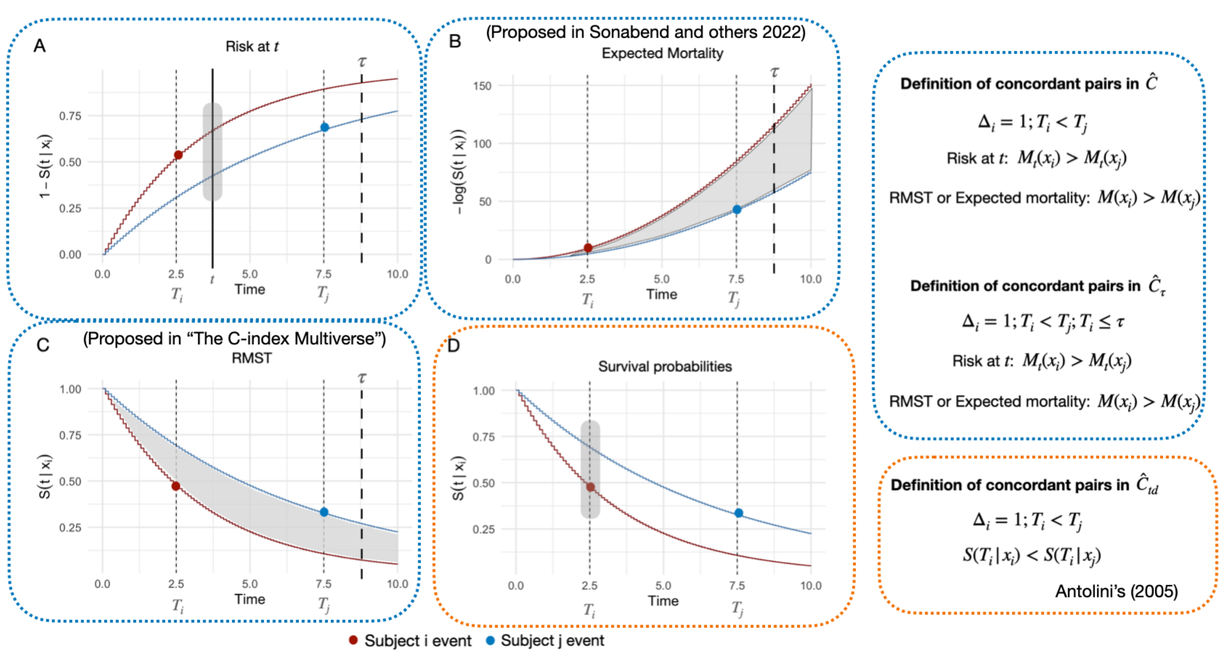
$$C_\tau = P(M(X_i) > M(X_j) \mid T_i < T_j, T_i < \tau)$$

Concordance estimator (Uno's 2011):

$$\widehat{C}_\tau = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} \Delta_i W_{ij} I(\tilde{T}_i < \tilde{T}_j, \tilde{T}_i < \tau) I(M(x_i) > M(x_j))}{\sum_{i=1}^{N}\sum_{j=1}^{N} \Delta_i W_{ij} I(\tilde{T}_i < \tilde{T}_j, \tilde{T}_i < \tau)}$$

$$W_{ij} = W_i W_j = \hat{G}(\tilde{T}_i)^{-2}$$

## Inputs to the C-index and C-index implementation mapping




Another source of ambiguity is the input given to the C-index.

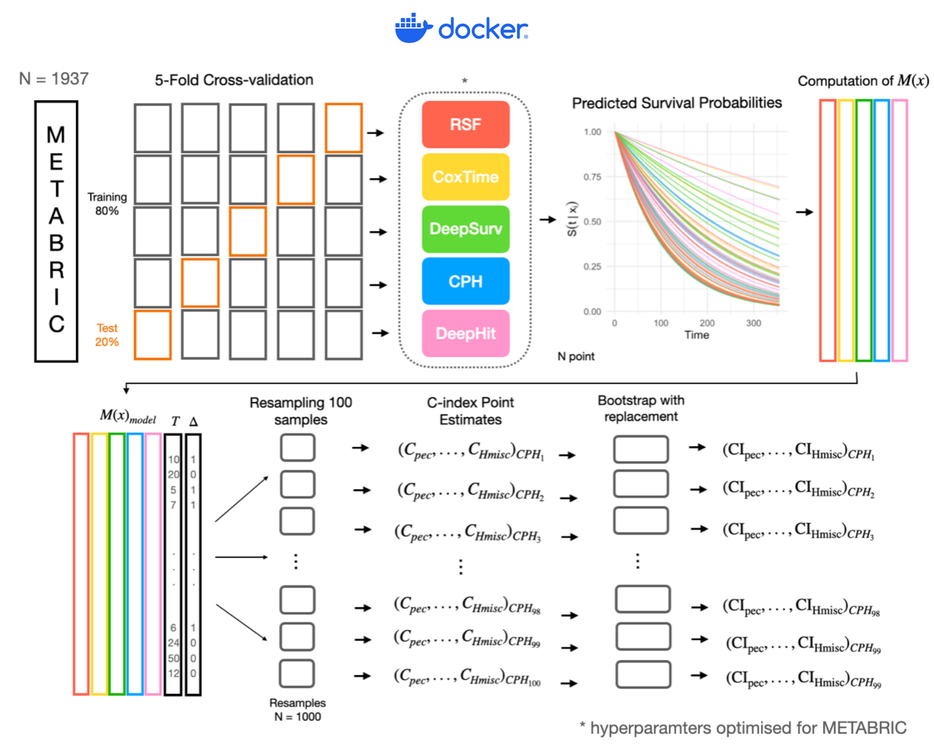
Mapping shows a C-index Multiverse

| Package/Function | Language | Tied Times | Tied Predictions | IPCW |
|---|---|---|---|---|
| **Implementations that estimate $C$** | | | | |
| `Hmisc / rcorr.cens` [1] | R | ✓ | ✓* | ✗ |
| `SurvMetrics / Cindex` [2] | R | ✓** | ✓** | ✗ |
| `Lifelines / concordance_index` [3] | python | ✓ | ✓ | ✗ |
| `pysurvival / concordance_index` [4] | python | ✓ | ✓* | ✓ |
| `Scikit-survival (sksurv) / concordance_index_censored` [5] | python | ✓ | ✓* | ✗ |
| **Implementations that estimate $C_\tau$** | | | | |
| `pec / cindex` [6] | R | ✓* | ✓* | ✓ |
| `survival / concordance` [7] | R | ✓ | ✓ | ✗* |
| `survC1 / Est.Cval` [8] | R | ✗ | ✓ | ✓ |
| `Scikit-survival (sksurv)/ concordance_index_ipcw` [5] | python | ✓ | ✓* | ✓ |
| **Implementations that estimate $C_{td}$** | | | | |
| `pycox / concordance_td` [9] | python | ✓*** | ✓*** | ✗ |

[1] Harrell Jr (2025), [2] Zhou *and others* (2023), [3] Davidson-Pilon (2019), [4] Fotso *and others* (2019), [5] Pölsterl (2020), [6] Gerds (2023), [7] Therneau (2024), [8] Uno (2021), [9] Kvamme and Borgan (2019)

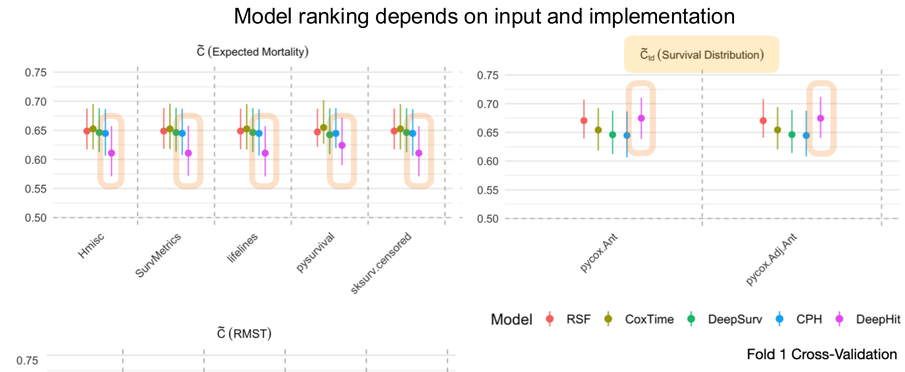
(*) Inclusion of ties and IPCW are a user-option.
(**) Considers tied times when $\Delta_i = \Delta_j = 1$.
(***) Considers tied times when $\Delta_i = 1$ & $\Delta_j = 0$.

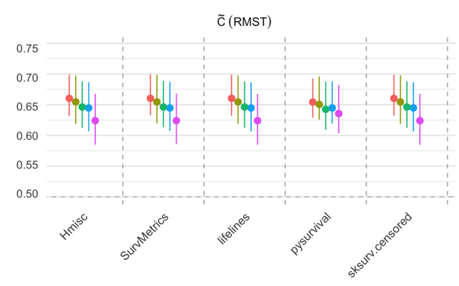
## Effects of the C-index Multiverse




### 1 Implementation differences impact model ranking

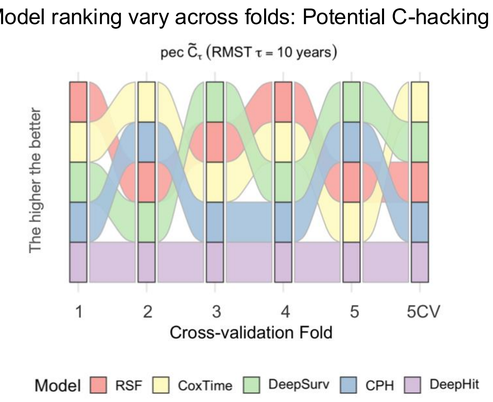
Model ranking depends on input and implementation

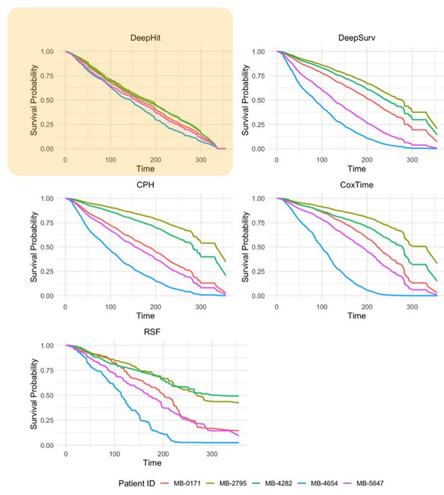


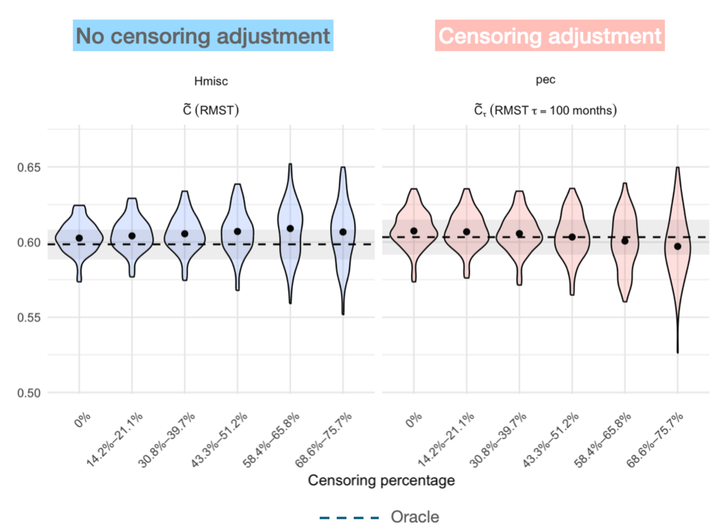

Solely looking at C-index, we miss potential miscalibrated models

Model ranking vary across folds: Potential C-hacking




### 2 Semi-synthetic METABRIC simulations: bias-variance trade-off when increasing censoring




**More in our paper:**

## Main takeaways

**It is not enough by saying "*we used the C-index*"…**

**The C-index has limitations, use it with other performance metrics such calibration or overall prediction error**

## What Harrell thinks of it?

f2harrell — Sep 2025

It looks like a good paper. An important reference from Brad Efron was omitted; I think he was one of the first to develop a fix for the dependence on the censoring distribution. I wish I had the reference handy.

While the *c*-index is a good measure of predictive discrimination for a single model, we need to quit using it to compare models. It's not sensitive enough for that purpose. I would prefer to emphasize pseudo adjusted R-square for all purposes.

## Acknowledgements

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